PROJECT

Forge Harness

Alloyed multi-model harnesses as a training data generator for small specialists.

Cybersecurity CTF   SFT + RL   Early results

01 — PROBLEM

No single model dominates everything

Each frontier model is strong in places and blind in others — and the blind spots don’t overlap.

Every model has blind spots. What if you never had to pick just one?

02 — THE CONCEPT

Alloying

Randomly alternate which model acts, turn by turn, inside one continuous identity-blind thread.

Pioneered by XBOW on security tasks.

Model A → one thread

Model B → no identity

68.8%
alloy

57.5%
solo

46.4%
solo

03 — THE INSIGHT

Alloying is used at inference to solve one task. We use it to generate training data.

alloy → trajectories → training

04 — WHAT WE COLLECT

Alloyed rollouts on CTF tasks

Reasoning chains no single model would have produced on its own.

Why CTF

Verifiable reward — the flag matches or it doesn't. Ground truth is free.

Where from

CTF-Dojo supplies the tasks and the execution environment. We started with Random-Crypto.

05 — TWO PATHS

Two paths from the trajectories

SFT

Distil directly

The small model imitates alloyed trajectories.

cheap & fast

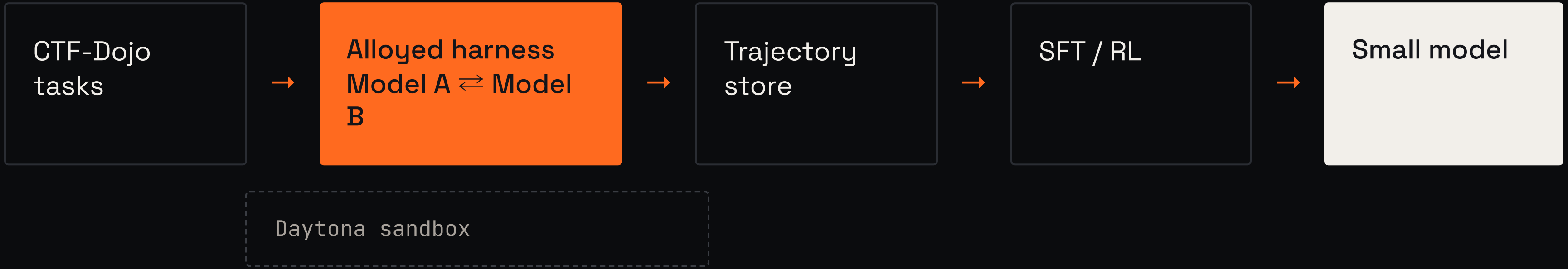
RL

Learn by rollout

Trajectories and outcomes shape the reward signal.

robust & generalises

Pipeline



The expensive multi-model alloy runs once, in data collection. Never at inference.

07 — STACK

Tech stack

Execution	Daytona sandboxes
Tasks	CTF-Dojo
Alloy	GLM 5.3 Flash · Sonnet 4.5 · Grok 4.6
Training	Unsloth (SFT + RL)

08 — EARLY RESULTS

Qwen 2.5 7B, trained on alloyed runs, roughly doubles on held-out Random-Crypto

one shot

20% → 37%

four shots

26% → 53%

Alloy of Sonnet 4.5, Grok 4.6 and GLM 5.3 Flash, one chosen at random each turn. Only flag-finding runs kept.

Student: Qwen 2.5 7B Instruct on a single 5090. Held-out set of 20 challenges it never trained on.

What’s new here

Pooling trajectories from several teachers is proven — CTF-Dojo reports the gains.

Training on turn-level alloyed trajectories, where teachers interleave inside one thread, hasn’t been done.

Thank you

Forge